

Project Initialization and Planning Phase

Date	15 Nov 2024
Team ID	739938
Project Title	AI Enabled Candidate Resume Screening
Maximum Marks	3 Marks

Project Proposal template

This project report gives information about how the different resumes are scanned through an machine, This AI helps the HR and recruiters to easily filters the skilled and the persons who are eligible to the job notification

Project Overview	
Objective	The main aim of AI-enabled candidate resume screening is to streamline the recruitment process by using artificial intelligence to analyze and filter resumes efficiently. some of the key objectives are to improve efficiency, enhance accuracy, eliminate bias, support data driven decisions, standardize screening, reduce costs, enhance candidate experience
Scope	The scope of AI-enabled candidate resume screening is broad, transforming recruitment with automation and precision. It includes parsing resumes to extract structured data, matching skills with job descriptions using Natural Language Processing (NLP), and ranking candidates based on relevance. AI systems can process large volumes of resumes quickly, reducing time-to-hire while maintaining accuracy
Problem Statement	
Description	The primary problem to be addressed regarding AI-enabled candidate resume screening is ensuring accuracy, fairness and transparency in the recruitment process while overcoming biases and technical limitations
Impact	AI-enabled candidate resume screening has significantly transformed the recruitment process by enhancing efficiency, scalability, and precision. It enables recruiters to quickly process large volumes of resumes, identifying top candidates through advanced algorithms that analyze skills, qualifications, and experience. By standardizing the

	screening process, AI reduces inconsistencies and potential human errors. However, challenges like algorithmic bias, over-reliance on keyword matching, and exclusion of non-standard resumes pose significant concerns. While AI improves efficiency, it cannot fully replicate human intuition, often struggling with subjective factors like creativity or cultural fit. To maximize its benefits, organizations must ensure ethical implementation, transparency, and a balanced approach combining AI with human judgment.
Proposed Solution	
Approach	AI-enabled candidate resume screening uses techniques like Natural Language Processing (NLP) to extract and analyze key details such as skills, education, and work experience. Machine Learning (ML) algorithms classify resumes, identify patterns, and improve over time by learning from recruiter feedback. Optical Character Recognition (OCR) converts resumes in various formats into machine-readable text, while predictive analytics evaluates candidates' fit for roles. Additionally, automated scoring ranks applicants based on relevance to job requirements, and bias detection algorithms help ensure fair assessments. These methods work together to streamline the recruitment process, making it faster and more efficient.
Key Features	AI-enabled candidate resume screening offers key features that enhance the recruitment process. It utilizes Natural Language Processing (NLP) to analyze resumes and extract important details like skills, qualifications, and work experience. Machine Learning (ML) algorithms classify and rank candidates based on relevance to job descriptions, while predictive analytics helps forecast candidate suitability and retention potential. The system incorporates Optical Character Recognition (OCR) to process resumes in various formats, ensuring compatibility and accessibility. Additionally, it provides automated scoring, bias detection to promote fair hiring, and seamless integration with Applicant Tracking Systems (ATS). These features enable faster, more accurate, and unbiased hiring decisions.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., intel i5 processor

Memory	RAM specifications	e.g., 8 GB
Storage	Disk space for data, models, and logs	e.g., 74kb
Software		
Frameworks	Python frameworks	e.g., Flask
Libraries	Additional libraries	e.g. Spacy, pyresparser, smtplib
Development Environment	IDE, version control	e.g., pycharm, Git
Data		
Resume	Source, size, format	e.g., canva, 74kb, pdf